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ITA0609

Exp-18: Implement Perceptron based IRIS classification

```
# Import required libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix
# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create the Perceptron model
model = Perceptron(random_state=42)
# Train the model
model.fit(X_train, y_train)
# Predict the test data
y_pred = model.predict(X_test)
# Display results
print("Actual Labels:")
print(y_test)
print("\nPredicted Labels:")
print(y_pred)
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)
```

Output:

```
Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Predicted Labels:
[1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0]

Accuracy: 63.33%

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0 11  0]]
```